

Who Bears the Burden of Bank Capital Regulation?

Buffers, Risk Weights, and the Supply of Credit

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Motivation

Why requirements exist: equity absorbs unexpected losses, so banks stay solvent in downturns—protecting insured deposits and financial stability.

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stability benefits of a higher requirement vs. the output cost of getting there

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But the regime is not an aggregate lever—it is deliberately differentiated:

- Risk weights: C&I loans $\omega_B = 1.00$ vs. mortgages $\omega_M = 0.50$
- Size tiers: surcharges at \$10B and \$50B asset thresholds (post-GFC)

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 - Size tiers: surcharges at \$10B and \$50B asset thresholds (post-GFC)
- Both margins presume regulators know *where a tightening binds*.

Who bears the burden of a capital requirement tightening?

- ▶ Which **sectors** absorb the contraction in credit?
- ▶ Which **banks** transmit it?
- ▶ What is the **aggregate cost**—and does design matter for it?

Paper in a Nutshell

Two facts (Call Reports, 4,800 banks, 1994–2019; identified shocks):

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A model disciplined by those facts:

- ▶ Reproduces the gradient at $1.23\times$, **untargeted** (though not the hump—simulated responses deepen without reverting)
- ▶ A +1pp tightening costs 1.4–2.4% of credit-market surplus

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The burden is organised by *distance to the frontier*, and the statutory margin explains just 1.8% of its variance.

Level of requirements $\Rightarrow$ aggregate cost. *Design* $\Rightarrow$ **incidence**.

What: how much capital each bank holds *above its own requirement*

- Call that gap the *voluntary buffer* $D_{i,t}$ — heterogeneous funding frictions γ_i generate it
- **This paper carries the whole distribution of $D_{i,t}$ across banks as a state variable**

Contribution

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- **This paper carries the whole distribution of $D_{i,t}$ across banks as a state variable**

Context: the closest quantitative models do not carry that distribution

- **Elenev, Landvoigt, and Van Nieuwerburgh (2021)** — one intermediary; failure risk i.i.d., survivors recapitalised each period $\Rightarrow$ incidence is *sector-level* by construction
- **Corbae and Erasmo (2021)** — two bank types; buffers disciplined at *two group means*, validated on a *size-sorted* regression

Source: Consolidated Reports of Condition and Income (Call Reports, FDIC)

- ▶ ~4,800 U.S. commercial banks, quarterly, 1994–2019
- ▶ Loans decomposed: real estate, C&I, consumption, credit card
- ▶ Macro controls: real GDP growth, federal funds rate (FRED)

¹Economic Growth, Regulatory Relief, and Consumer Protection Act (2018), which raised the automatic enhanced-prudential-standards threshold from \$50B to \$250B. The \$50B tier governs almost the whole 1994–2019 sample.

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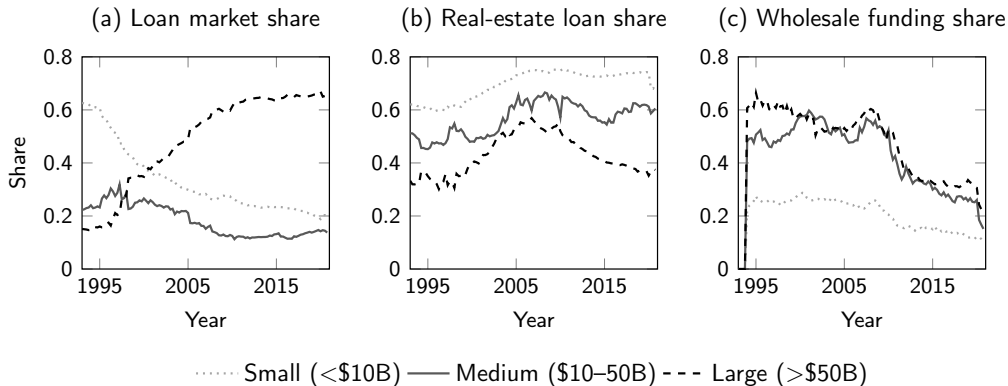
Size cohorts (Dodd-Frank/EGRRCPA¹ thresholds):

- ▶ Community, $< \$10\text{B}$ — baseline κ
- ▶ Regional, $\$10\text{--}50\text{B}$ — $\kappa + 1.5\text{pp}$ (Dodd-Frank)
- ▶ SIFI, $> \$50\text{B}$ — $\kappa + 2.5\text{pp}$ (G-SIB surcharge)

Regulatory wedge

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Market Concentration and Asset Specialisation



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$r_{i,t}$ is the bank's total risk-based capital ratio (regulatory capital / RWA, Call Reports)

$\kappa_{i,t}$ is bank i 's own statutory requirement—baseline, +1.5pp (DFA) or +2.5pp (G-SIB)

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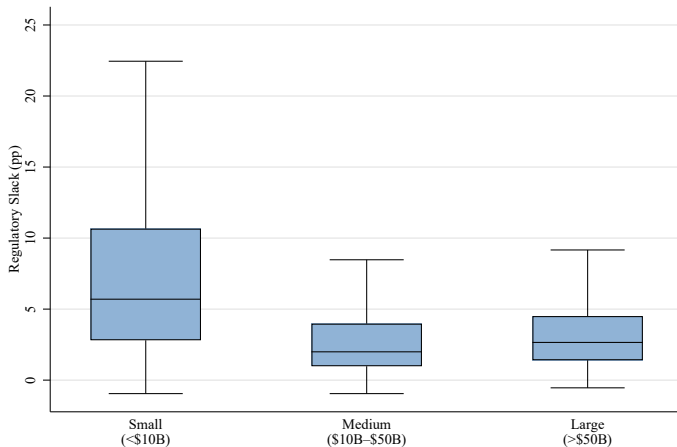
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Funding Capacity $\equiv$ (core retail + NIB deposits) / total liabilities

- ▶ A one-standard-deviation higher core-deposit share (≈ 0.10) goes with **2.0–2.6pp** less regulatory slack — about 40% of the median buffer of 5.7pp
- ▶ Strong comparing *different* banks, near-zero for a *given* bank over time — a funding model is structural, not a quarterly choice

Buffer Heterogeneity Cuts Across Size Tiers



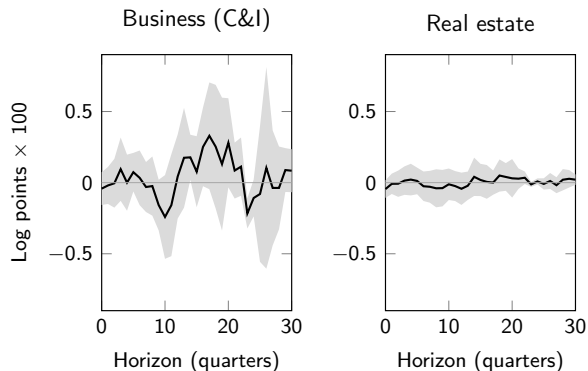
Source: US Call Reports, 1993Q2–2019Q4.
Regulatory slack: equity-to-RWA ratio minus effective requirement (κ_{eff}), winsorized at 1/99.

Fact 1: Sectoral Asymmetry in Credit Responses

Identification: Narratively identified capital-requirement shocks (Drechsel and Miura, 2025);
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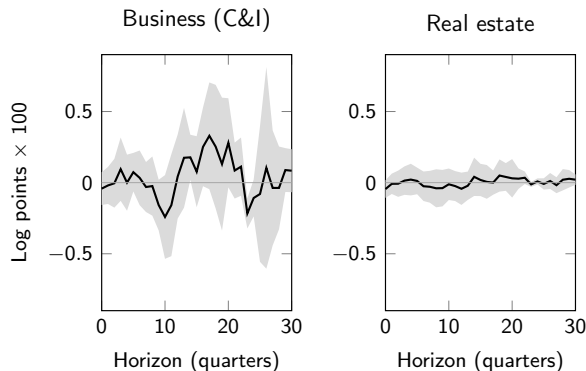
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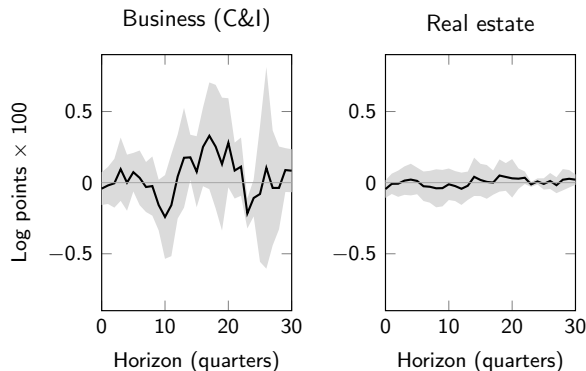


Asset Substitution Margin

When the constraint binds, banks shed high- ω assets first: a dollar of C&I frees the most required capital.

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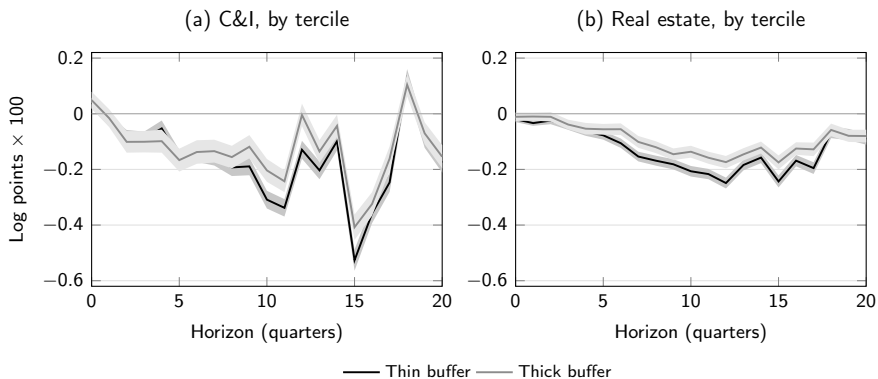
When the constraint binds, banks shed high- ω assets first: a dollar of C&I frees the most required capital.

Scale Deleveraging Multiplier

$$\frac{\partial RWA}{\partial A^B} = \omega_B > \omega_M = \frac{\partial RWA}{\partial A^M}$$

Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut.

Fact 2: The Buffer Gradient



Banks sorted by lagged regulatory slack $D_{i,t-1}$.

Thin-buffer banks contract $1.32\times$ more in C&I; the model reproduces $1.23\times$, **untargeted**.

Details

Precision

What the Buffer Gradient Identifies

A capital constraint alone cannot produce it.

- ▶ With regulation entering only as a constraint, an unconstrained bank's first-order condition contains no κ : $\partial A_i^j / \partial \kappa = 0$ *exactly*

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- ▶ $c = 0.148$ (not risk-weighted), $z = 0.021$ (risk-weighted)
set on the two sectoral aggregates, both buffer gradients **untargeted**

Model

Facts recap

Model: Environment

Dynamic, partial equilibrium, heterogeneous-agent.

- ▶ Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- ▶ Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$

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- ▶ Core deposits $d_{i,C} \leq \bar{d}_{i,C}$, plus wholesale $d_{i,t}^W$ at quadratic cost $\frac{\gamma_i}{2} (d_{i,t}^W)^2$
- ▶ $\gamma_{\text{small}} \gg \gamma_{\text{large}} \rightarrow 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from

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- ▶ One aggregate disturbance: κ_t follows an AR(1); bank i faces $\kappa_{i,t} = \kappa_t +$ its statutory tier add-on
- ▶ Two occasionally binding constraints: regulatory capital, and dividend non-negativity

The Dynamic Bank Problem (I): The Problem

Banks choose lending and dividends to maximise the discounted dividend stream:

$$V(e_{i,t}, A_{i,t-1}^M, A_{i,t-1}^B; \Omega_t) = \max \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^{\tau} c_{i,t+\tau}$$

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Subject to the balance-sheet identity — wholesale funding is *residual*, not chosen:

$$d_{i,t}^W = A_{i,t}^M + A_{i,t}^B - e_{i,t} - \bar{d}_{i,C}$$

and to the law of motion for equity, which carries returns, costs and the levy into $e_{i,t+1}$.

Equity law of motion

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Equity law of motion

and to two occasionally binding constraints:

$$\underbrace{\omega_M A_{i,t}^M + \omega_B A_{i,t}^B}_{\text{regulatory capital}} \leq \frac{e_{i,t}}{\kappa_{i,t}}, \quad \underbrace{c_{i,t}}_{\text{no equity issuance}} \geq 0$$

The Dynamic Bank Problem (II): The Euler Equation

Optimal lending in sector j satisfies:

$$\beta \mathbb{E}_t \left[(1 + \mu_{i,t+1}) \left(r_{j,t} - \varepsilon_{i,t+1}^j - \psi_j A_{i,t}^j - r_{W,t} - \gamma_i d_{i,t}^W - \phi_{i,j}(\kappa_{i,t}) \right) \right] = \lambda_{i,t} \omega_j$$

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- ▶ $\lambda_{i,t}$: shadow cost of the capital constraint — zero when the bank is away from the frontier
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- ▶ $\phi_{i,j}$: per-unit cost of regulatory capital, paid *whether or not* the constraint binds (bank-specific via $\bar{\kappa}_i$)
- ▶ $\omega_B > \omega_M \Rightarrow$ higher regulatory cost per unit C&I $\Rightarrow$ C&I contracts more

Where these terms come from

Neural-Network Solution

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- **Structural loss:** squared Euler residuals + soft KT penalty
- **Market clearing:** imposed analytically via closed-form equilibrium rate
- **Architecture:** 4 hidden layers $\times$ 256 nodes, Swish activation
- **Optimizer:** AdamW with cosine annealing

Network

Method

Non-convexity

Solution accuracy

What the policy pins down

SMM Estimation Strategy

Estimate: $\theta = (\rho_\kappa, \sigma_\kappa)$ — persistence and volatility of the capital-requirement process

Targets: 16 within-bank LP coefficients

C&I and RE responses at $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$

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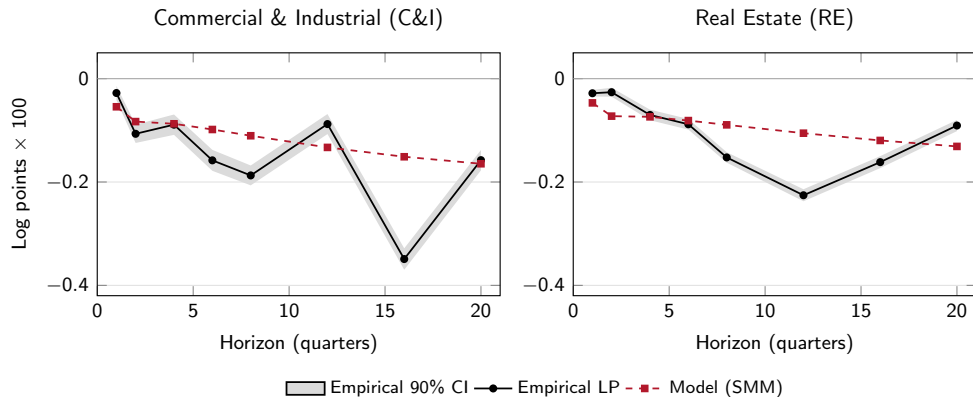
Identification, in words:

- ▶ $\rho_\kappa \leftarrow$ how slowly the IRF decays at medium horizons
- ▶ $\sigma_\kappa \leftarrow$ how large the impact response is
- ▶ Two parameters, sixteen moments — the process is over-identified.

Weighting and algorithm

Calibration

Estimation: Model vs. Data

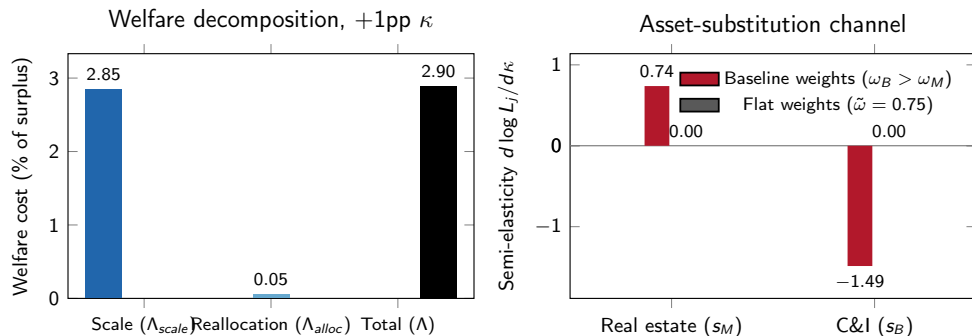


$\hat{\rho}_{\kappa} = 0.9987$ (s.e. 0.004), $\hat{\sigma}_{\kappa} = 0.00040$ (s.e. 0.00002) — interior, within 1 s.e. of a unit root.

Moment fit

Robustness

Gross Cost and Decomposition



Left: CEV decomposition. Right: constrained-regime semi-elasticities.

Counterfactuals: neither a uniform $\tilde{\kappa}$ nor flattened risk weights changes the aggregate burden.

Counterfactuals

Details

CEV framework

Magnitudes in context

Who Bears It? Incidence Across Banks

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- ▶ **Thin-buffer banks:** 32% of lending, 34% of the burden — concentration 1.06
- ▶ **Thick-buffer banks:** 35% of lending, 33% of the burden — concentration 0.94

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Thin-buffer banks cut 13% more *per dollar lent*, but they are smaller — so the burden is disproportionate per unit of credit, not concentrated on a small group.

Who bears the burden of bank capital regulation?

- ▶ **Which sectors:** risk weights concentrate the contraction in C&I ($1.31\times$ vs real estate)
- ▶ **Which banks:** distance to the frontier, not size ($1.32\times$ in the data, $1.23\times$ untargeted in the model)

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The level of requirements sets the aggregate cost (1.4–2.4% of credit-market surplus per +1pp); their design moves only who pays it.

Gross transition costs excluding stability benefits.

GE channels (borrower substitution to non-banks, loan-rate adjustment) would attenuate the real-side impact; aggregate-demand feedback could amplify it.

Thank you

Who Bears the Burden of Bank Capital Regulation?

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Interpreting the Sectoral Asymmetry

Why C&I contracts more than RE:

► **Asset Substitution Margin:** when the constraint binds, banks shed high- ω assets (C&I, $\omega_B = 1$) to free the most required capital per dollar cut

► **Scale Deleveraging Multiplier:** $\partial RWA / \partial A^B = \omega_B > \omega_M = \partial RWA / \partial A^M$

Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut

► **Consumer credit unresponsive:** high interest margins provide sufficient cash-flow buffers to absorb the regulatory cost without volume rationing

Result: the sectoral composition of the contraction is determined by the risk-weight schedule, not by loan demand.

SMM Moment Fit

Sector	Horizon h	$\hat{b}_h$ (data)	s.e.	b_h (model)	t
C&I	1	-0.028	0.008	-0.054	+3.20
	2	-0.107	0.011	-0.083	-2.20
	4	-0.089	0.012	-0.087	-0.14
	6	-0.158	0.012	-0.098	-4.85
	8	-0.187	0.012	-0.110	-6.57
	12	-0.088	0.012	-0.133	+3.84
	16	-0.349	0.013	-0.151	-15.73
	20	-0.158	0.012	-0.165	+0.58
RE	1	-0.028	0.004	-0.047	+4.87
	2	-0.026	0.005	-0.073	+8.80
	4	-0.070	0.006	-0.074	+0.59
	6	-0.088	0.006	-0.081	-1.15
	8	-0.152	0.006	-0.089	-10.29
	12	-0.226	0.007	-0.106	-17.02
	16	-0.162	0.007	-0.120	-5.97
	20	-0.091	0.007	-0.131	+6.16
Hansen $J = 1822.15$, $\text{dof} = 14$, $\chi^2_{0.05} = 23.68$, $p = 0.000$					

Signs correct for all 16 moments; sectoral ordering reproduced at every horizon. $J = 1822$ (14 d.f.) reflects the high precision of the bank panel—the economically relevant misfit is in the trough depths.

Identification: Capital-Requirement Shocks

Shock series: Drechsel and Miura (2025)

- Narrative + sign-restriction approach
- Captures unanticipated changes in the capital-requirement stance
- Available 2002Q1–2023Q2; estimation sample capped at 2019Q4

Local Projection specification:

$$x_{t+h} - x_{t-1} = \alpha_h + \beta_h \text{shock}_t^{\text{Cap}} + \sum_{p=1}^6 \psi_{h,p} x_{t-p} + \varepsilon_{t+h}$$

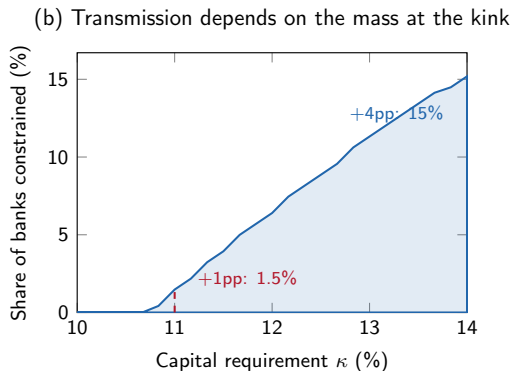
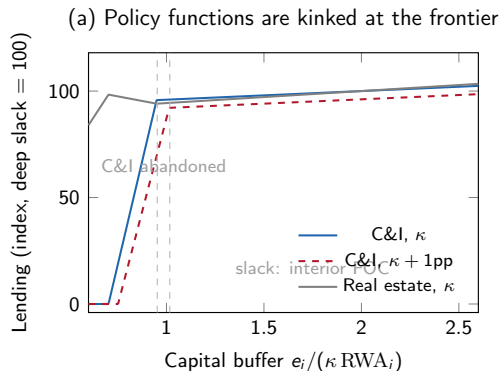
- ▶ **Aggregate:** OLS, Newey–West (lag $h + 6$)
- ▶ **Bank-level:** bank FE, clustered by cert
- ▶ Consumer credit placebo: no response $\Rightarrow$ supply, not demand

The Buffer Gradient: Details

- ▶ Thin-buffer banks cut C&I $1.32\times$ more than thick-buffer banks (90% CI [1.01, 1.62]); $1.41\times$ in real estate, $1.37\times$ pooled [1.13, 1.61] — from a joint cluster-robust covariance across the separate tercile regressions
- ▶ The risk-weight mechanism implies the *same* gradient in both sectors, and the restriction is not rejected: difference $+0.090$ (s.e. 0.186), $t = 0.48$
- ▶ Freezing tercile membership at each bank's pre-sample average slack **reverses** the gradient, to 0.67 in C&I — it is the buffer a bank holds when the shock arrives that matters, not a fixed bank type

Horizon profile, cohort power, and the continuous-interaction specification: see backup.

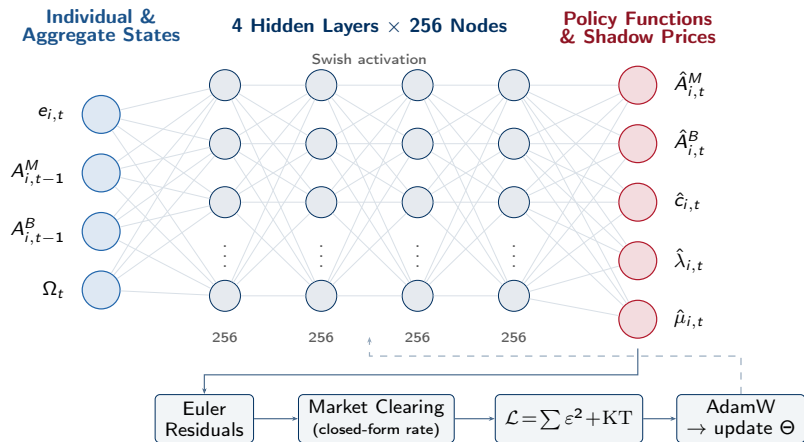
Why a Global Non-Linear Method



Both panels are the *exact* static solution — the problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime.

- ▶ The capital constraint binds occasionally, so the policy function is **kinked**: left of the frontier lending is pinned by equity, right of it the interior FOC holds and lending is flat in equity
- ▶ A +1pp tightening moves the kink *right* — banks that were unconstrained become constrained
- ▶ So the aggregate response is not a fixed elasticity: it depends on the **mass sitting near the kink**. +1pp constrains 1.5% of banks, +4pp constrains 15%

Neural-Network Architecture



Training: $\sim 6h$ on NVIDIA A100 · Cosine annealing schedule · Euler errors $< 10^{-3}$ across state space

Back

Gross Cost: Numbers

Experiment	Quantity	Value
<i>CEV of +1pp κ tightening (eqs. (??)–(??))</i>		
Total welfare cost	Λ	1.421%
scale channel	Λ_{scale}	1.415%
allocation channel	Λ_{alloc}	0.004%
additivity check	$\Lambda_{scale} + \Lambda_{alloc}$	1.419%
Credit contraction (total / RE / C&I)	ΔL	-3.09% / -3.02% / -3.50%
<i>CEV from the franchise value V (discounted dividends)</i>		
Total welfare cost	Λ_V	2.404%
scale channel	$\Lambda_{V,scale}$	2.392%
allocation channel	$\Lambda_{V,alloc}$	0.026%
additivity check	sum	2.419%
<i>CF1: uniform vs. size-differentiated κ (eq. (??))</i>		
Differentiated rule	Λ	1.236%
Uniform $\tilde{\kappa} = 0.1246$	Λ	1.239%
<i>CF2: flat risk weights $\tilde{\omega} = 0.75$ (eq. (??))</i>		
Baseline sectoral asymmetry	$s_B - s_M$	-2.224
Flat-weight sectoral asymmetry	$s_B - s_M$	-0.000

Credit-market surplus: $\Lambda = 1.4\%$. Franchise value: $\Lambda_V = 2.4\%$. Scale channel accounts for $>99\%$ of the cost. Magnitudes are *directional*: robust in sign and decomposition, sensitive to the solved policy. The cross-sectional gradient is not.

Policy Counterfactuals

CF1: Uniform vs. size-differentiated κ

- ▶ Replace the step function with a uniform $\tilde{\kappa} = 0.1246$ raising the same aggregate capital
- ▶ Gross cost: 2.680% vs. 2.682% $\Rightarrow$ size-tailoring redistributes but barely reduces the aggregate burden

CF2: Risk-weight flattening

- ▶ Set $\tilde{\omega}_B = \tilde{\omega}_M = 0.75$ (average), hold κ fixed
- ▶ Sectoral asymmetry $s_B - s_M$ collapses from -2.22 to 0.00
- ▶ Aggregate credit contraction essentially unchanged

Level of requirements $\Rightarrow$ aggregate cost.
Design (risk weights, size tiers) $\Rightarrow$ incidence.

Backup: Counterfactual Definitions

CF1 — uniform vs. size-differentiated requirements. Replace the statutory step function $\kappa_i(\text{Scale}_i)$ with the single rate $\tilde{\kappa}$ that raises the *same aggregate capital* in the stationary distribution μ^* :

$$\tilde{\kappa} \text{ s.t. } \int \tilde{\kappa} e_i \, d\mu^* = \int \kappa_i(\text{Scale}_i) e_i \, d\mu^*.$$

So the experiment holds the *level* of regulation fixed and varies only its *design*: any difference in Λ is pure incidence.

CF2 — flat risk weights. Set $\omega_M = \omega_B = \tilde{\omega} = 0.75$, removing the risk-weight wedge. The test is whether the two sectors then deleverage proportionally:

$$\frac{\partial \tilde{L}_{t+h}^B}{\partial \varepsilon_t^\kappa} \approx \frac{\partial \tilde{L}_{t+h}^M}{\partial \varepsilon_t^\kappa} \quad \forall h.$$

	Λ	sectoral asymmetry $s_B - s_M$
Differentiated (baseline)	1.236%	-2.224
Uniform $\tilde{\kappa} = 0.1246$	1.239%	—
Flat weights $\tilde{\omega} = 0.75$	—	-0.000

Design redistributes the burden; it barely moves the aggregate. Flattening the weights eliminates the sectoral asymmetry entirely.

Backup: Related Literature

Quasi-experimental credit supply

Aiyar, Calomiris, and Wieladek (2016), Gropp, Guettler, and Saadi (2020), De Jonghe, Dewachter, and Ongena (2020), Jimenez et al. (2017), Conti, Nobili, and Signoretti (2023)

Here: the same identified shock is traced across sectors *and* across the buffer distribution, then mapped into a structural cost.

Deposit franchise and regulatory incidence

Drechsler, Savov, and Schnabl (2017)

The deposit franchise shapes not only monetary transmission but the incidence of *capital* regulation: what disciplines the buffer is the *marginal* funding source.

Quantitative macro-banking

Corbae and Erasmo (2021), Begenau (2020), Elenev, Landvoigt, and Van Nieuwerburgh (2021), Mendicino et al. (2020)

Backup: Reading c and z

$$\phi_{i,j}(\kappa) = (c + z\omega_j) \max(\kappa - \bar{\kappa}_i, 0)$$

- ▶ $z = 0.021$ is **risk-weighted**: a Modigliani–Miller premium, $\approx 10\%$ annualised equity–debt spread
- ▶ $c = 0.148$ is **not** risk-weighted and seven times larger — what a **leverage requirement** imposes. U.S. banks face one; the model's constraint is risk-based only
- ▶ Both are set on the two sectoral aggregates; *both* buffer gradients are untargeted

The leverage-requirement reading is a conjecture about the *source* of the non-risk-weighted term. The data identify its *magnitude* either way.

Backup: SMM Weighting and Algorithm

Weighting matrix

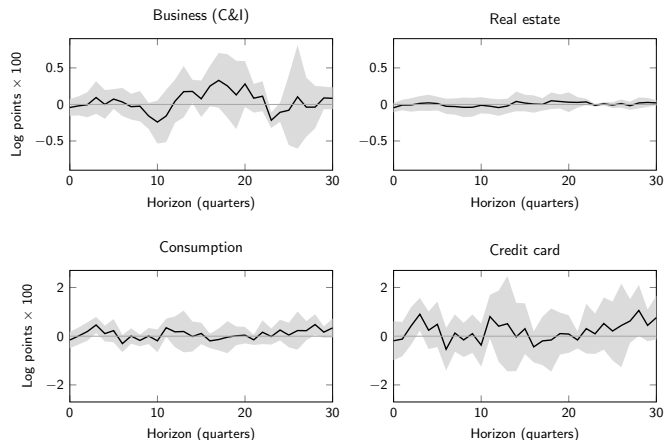
$W = \hat{\Sigma}^{-1}$, block-diagonal, estimated by cluster bootstrap of the bank-level local projections (100 replications)

Algorithm — for each candidate $\theta = (\rho_{\kappa}, \sigma_{\kappa})$

- 1 Re-solve the model globally with the neural network
- 2 Simulate a panel impulse path under the candidate process
- 3 Compute model-implied LP coefficients on the simulated panel
- 4 Evaluate $(\hat{m} - m(\theta))'W(\hat{m} - m(\theta))$

Each evaluation requires a full re-solve, which is what makes the global solution method the binding cost of estimation.

Backup: All Four Loan Categories



C&I contracts most, real estate less, and consumption and credit-card lending do not respond at any horizon. The two unresponsive categories are the demand placebo: they are fully risk-weighted but funded against high interest margins, so a capital tightening need not be absorbed through volume.

Backup: Summary of Empirical Facts

Fact	Finding
1. Sectoral asymmetry	C&I contracts $\sim$ a third more than RE within banks (risk-weight wedge)
2. Buffer gradient	Thin-buffer banks cut $1.32\times$ more; with terciles frozen at pre-sample slack the gradient <i>reverses</i> (0.67)
3. Placebo	Consumption and credit-card lending unresponsive at all horizons
4. Robustness	Results stable to 2023Q2 extension, pandemic controls

Backup: Where the Euler Terms Come From

Law of motion for equity. Every term in the Euler equation is a derivative of this line.

$$e_{i,t+1} = (1 - \delta_e) e_{i,t} + \sum_{j \in \{M, B\}} \left[(r_{j,t} - \varepsilon_{i,t+1}^j) A_{i,t}^j - \frac{\psi_j}{2} (A_{i,t}^j)^2 - \phi_{i,j}(\kappa_{i,t}) A_{i,t}^j \right] \\ - r_{d,t} \bar{d}_{i,C} - r_{W,t} d_{i,t}^W - \frac{\gamma_i}{2} (d_{i,t}^W)^2 - c_{i,t} - \tau_t e_{i,t}$$

Wholesale funding is residual, not chosen: $d_{i,t}^W = A_{i,t}^M + A_{i,t}^B - e_{i,t} - \bar{d}_{i,C}$

$\varepsilon_{i,t+1}^j$: idiosyncratic credit loss, realised after lending, so the net return per dollar is $r_{j,t} - \varepsilon_{i,t+1}^j$

ψ_j : quadratic operating cost — marginal cost $\psi_j A_{i,t}^j$ rises with portfolio size, which pins the long-run sectoral mix

$r_{d,t}$, $r_{W,t}$: retail deposit and wholesale funding rates

δ_e : quarterly equity depreciation (operating costs, franchise decay)

τ_t : systemic insurance levy on equity, funding entrant capital injections

Backup: Slack on Funding Capacity

Dep. var: Slack (pp)	(1) Pooled	(2) Between	(3) Within
Funding Capacity	−19.44*** (1.73)	−25.97*** (1.95)	−2.27** (1.05)
Log Assets	−2.80*** (0.09)	−2.70*** (0.10)	−6.27*** (0.16)
Quarter FE / Bank FE	Yes / No	—	Yes / Yes
<i>N</i>	935,234	16,117	935,056

The between estimate (−25.97) is the cross-sectional regularity; the within estimate (−2.27) is small because the funding mix barely moves at quarterly frequency. Size enters separately and negatively throughout, so this is not size in disguise.

Stylized Model: Environment

Static, partial equilibrium. Continuum of risk-neutral banks $i \in [0, 1]$.

- ▶ Two asset classes: mortgages (A_i^M , $\omega_M = 0.50$) and C&I (A_i^B , $\omega_B = 1.00$)
- ▶ Liabilities: core deposits $d_{i,C} \leq \bar{d}_{i,C}$ (cheap, capped) + wholesale $d_{i,W}$ (quadratic friction $\frac{\gamma_i}{2} d_{i,W}^2$)
- ▶ $\gamma_{\text{small}} \gg \gamma_{\text{large}} \rightarrow 0$: small banks face steep wholesale costs
- ▶ Regulatory constraint: $e_i \geq \kappa(\omega_M A_i^M + \omega_B A_i^B)$

Capital buffer:

$$D_i = e_i - \kappa(\omega_M A_i^M + \omega_B A_i^B) \geq 0$$

Banks with high $\gamma_i \Rightarrow$ large endogenous D_i (self-insurance).

First-Order Conditions and the Sectoral Channel

FOC for asset class $j \in \{M, B\}$:

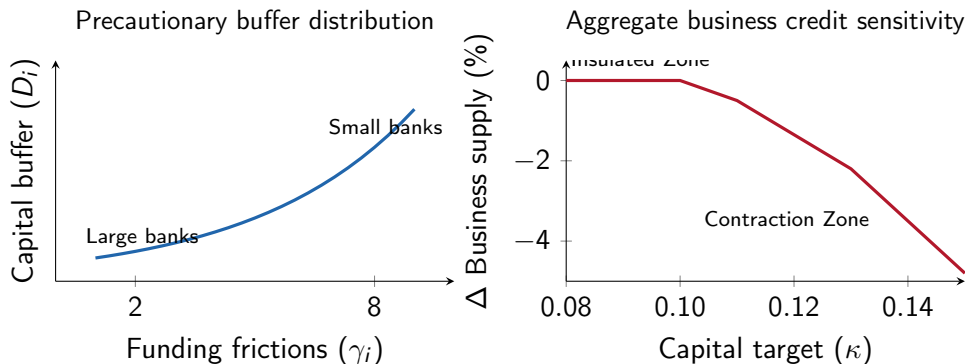
$$r_j - r_W - \psi_j A_i^j - \gamma_i (A_i^M + A_i^B - e_i - d_{i,C}) = \lambda_i \kappa \omega_j$$

When $\lambda_i > 0$ (constraint binds):

- ▶ **Asset Substitution Margin:** $|\partial A_i^B / \partial \kappa| > |\partial A_i^M / \partial \kappa|$ because $\omega_B > \omega_M$
- ▶ **Scale Deleveraging Multiplier:** $\partial RWA / \partial A^B = \omega_B = 1 > 0.5 = \omega_M \Rightarrow$ cutting C&I frees 2× the regulatory capital per dollar

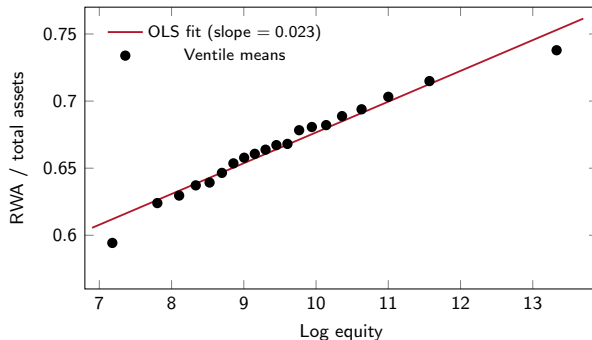
Cross-section: for low- γ banks, λ_i stays low $\Rightarrow$ small response. For high- γ banks whose buffer is breached, λ_i spikes $\Rightarrow$ acute, non-linear contraction in C&I.

The Insulated Zone



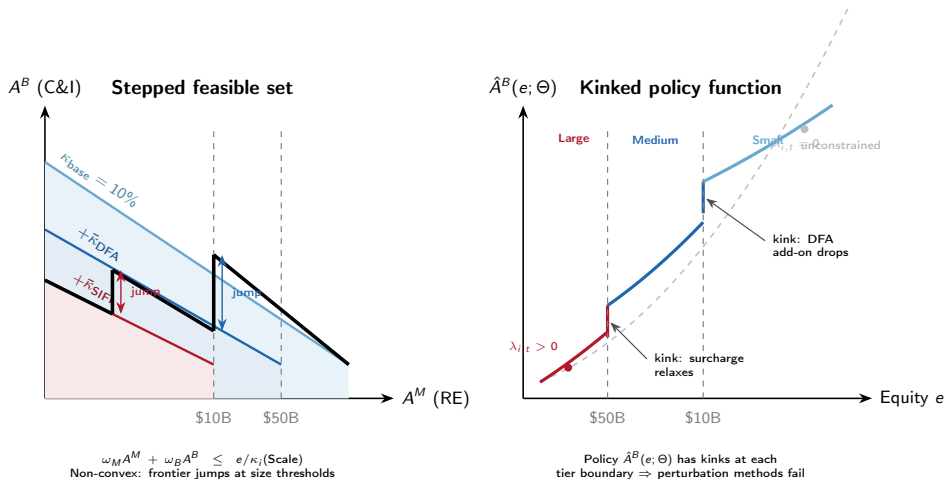
When shocks are small-to-moderate, well-buffered banks absorb them internally. Only shocks large enough to breach the buffer trigger non-linear credit contractions.

The Regulatory Wedge



RWA/total-assets ratio rises with bank equity $\Rightarrow$ larger banks hold higher average risk weights; portfolio composition drives the regulatory position of small banks.

Why Neural Networks? The Non-Convexity Challenge



Size-tiered surcharges create a stepped feasible set (left) and kinks in optimal policy (right). Standard perturbation around the steady state cannot capture these discontinuities—motivating a global neural-network solution.

Calibration

Parameter	Description	Value	Source
β	Discount factor	0.99	Standard
δ_e	Equity depreciation	0.001	Stationarity
ω_M, ω_B	Risk weights	0.50, 1.00	Basel III
κ_{base}	PCA minimum	10%	Regulatory
$\bar{\kappa}_{\text{DFA}}$	DFA add-on	1.5%	Enhanced prudential
$\bar{\kappa}_{\text{SIFI}}$	SIFI add-on	2.5%	G-SIB surcharge
γ_{small}	Wholesale friction	0.077	Funding structure
ψ_M	RE adjustment cost	0.007	RE share target
ψ_B	C&I adjustment cost	0.05	Preset
η_M, η_B	Demand slopes	external	Lit. semi-elasticities
$\sigma_{\varepsilon, M}$	Idiosync. shock (RE)	0.001	Charge-off data
$\sigma_{\varepsilon, B}$	Idiosync. shock (C&I)	0.004	Charge-off data

Consumption-equivalent variation (CEV):

$$V^{\kappa}(\mu^*; \kappa^* + 0.01) = V^*((1 - \Lambda)\mu^*; \kappa^*)$$

$\Lambda > 0$ = fraction of steady-state surplus households would forgo to avoid the tightening.

Decomposition:

$$\Lambda = \underbrace{\Lambda^{\text{alloc}}}_{\text{sectoral distortion}} + \underbrace{\Lambda^{\text{scale}}}_{\text{credit contraction}}$$

- ▶ Two metrics: credit-market surplus and franchise value (discounted dividends)
- ▶ **Caveat:** gross cost—no stability benefits, partial equilibrium

Backup: Robustness to Sample Extension

Re-estimating small-cohort LPs on the full shock series (2002Q1–2023Q2):

Treatment	Mean C&I response	Mean RE response
Baseline (2002–2019)	−0.145%	−0.105%
Pandemic dummy $\times$ shock	−0.142%	−0.106%
Exclude pandemic shocks	−0.142%	−0.106%

Target moments are essentially unchanged.

[Sensitivity to \$\psi_B\$](#)

[Back](#)

Backup: Sensitivity to ψ_B

Constrained-regime sectoral asymmetry $s_B - s_M$ across the admissible range of $\psi_B \in (0.014, 0.142)$:

ψ_B	$s_B - s_M$
0.02	-3.5
0.05 (baseline)	-2.2
0.10	-0.8
Flat weights (any ψ_B)	0.0

The asymmetry is a risk-weight phenomenon; ψ_B governs only its amplitude.

Backup: Precision of the Buffer Gradient

Terciles are estimated as separate local projections, so the *difference* needs a joint covariance: a bank contributes scores to several tercile blocks over its life, and the twenty horizons use overlapping windows.

	Thin	Thick	Ratio (90% CI)
C&I	−0.1718	−0.1307	1.32 [1.01, 1.62]
Real estate	−0.1354	−0.0964	1.41 [1.14, 1.67]
Pooled	—	—	1.37 [1.13, 1.61]

The cross-sector restriction is not rejected. The risk-weight mechanism implies the same gradient in both sectors: measured difference +0.090 (s.e. 0.186), $t = 0.48$, across 9,801 banks.

[Back](#)

Backup: What the Solved Policy Does and Does Not Pin Down

The model is solved by training a policy network. What that choice affects is not what an earlier version of this slide claimed.

Stable across solved policies. The welfare cost. With the cost of regulatory capital active, Λ lies between 1.16% and 1.42% across solved lending from $0.21\times$ to $1.05\times$ the calibration target — it tracks the *proportional* contraction ($\approx -3\%$), not the level.

Not stable. The impulse-response magnitudes, and the buffer gradient when it is recomputed *from* a solved policy rather than from the static problem.

So the gradient we report is the exact one. The static problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime: C&I 1.32 against 1.32 in the data. The trained policy returns 1.02–1.06 on the same calibration — it smooths the kink at the frontier, where its residuals are largest — and neither added capacity nor a four-fold longer schedule closes the gap.

What collapses Λ to 0.1% is not a low solved level but *removing* $\phi_{i,j}$: without it total credit moves -0.25% rather than -3.1% . That is a different model, not a numerical sensitivity — and it is the sharpest evidence that $\phi_{i,j}$, not the capital constraint, generates the welfare cost.

Backup: Sensitivity to the Demand Calibration

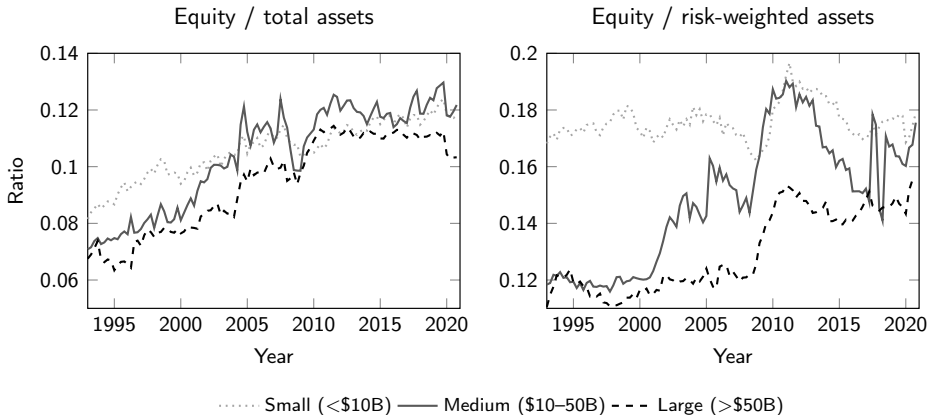
η_j enters neither the bank's problem nor the reduced-form estimates, so the lending responses are *exactly* invariant to it. The welfare *level* is not, because the surplus integral is taken under the demand curve.

S_M	S_B	Λ
32.5 (implemented)	14.8	1.42%
16.3	7.4	0.95%
2.0 (cited targets)	4.0	0.27%

The externally cited estimates measure objects adjacent to a loan-demand schedule — a house-price elasticity, and a loan-*granting* probability against the policy rate — rather than the schedule itself.

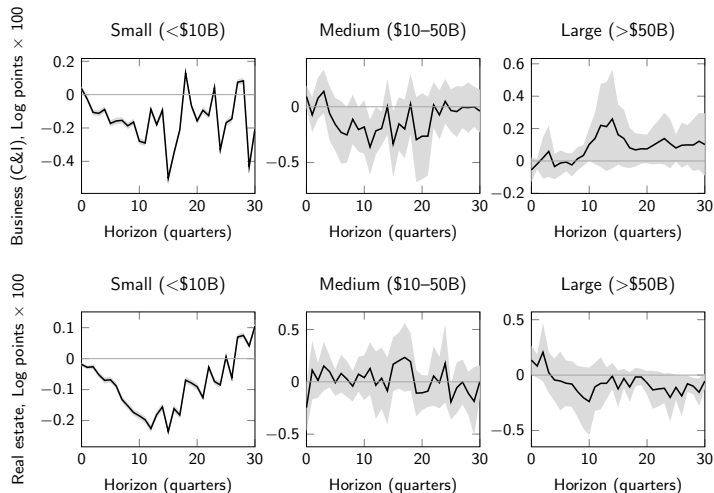
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Backup: Leverage vs. Regulatory Capital Ratios



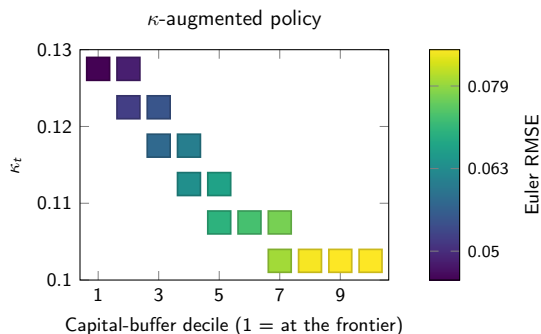
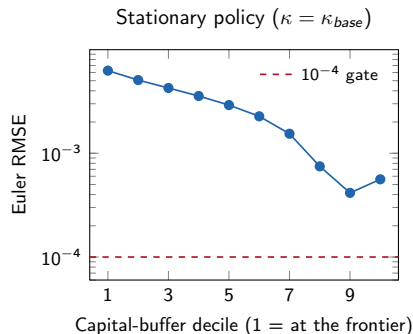
Small banks appear more capitalised on an unweighted basis, but the advantage compresses under risk-weighting.

Backup: Size Cohort Impulse Responses



Small-bank panel precisely estimated; large-bank cohort (≈ 40 institutions) is not. The buffer distribution, not size per se, drives the cross-section.

Backup: Solution Accuracy



All deciles below 10^{-2} . Frontier spike reflects the kink in policy functions at the occasionally binding constraint.

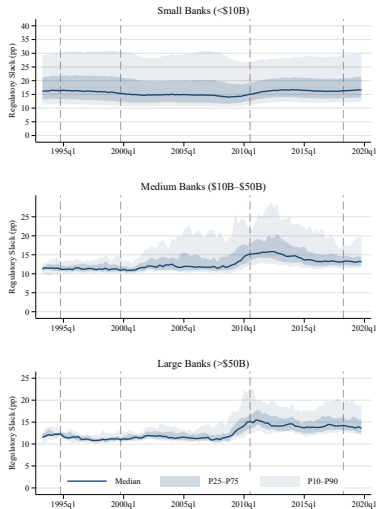
Train vs. test

Ergodic coverage

Back

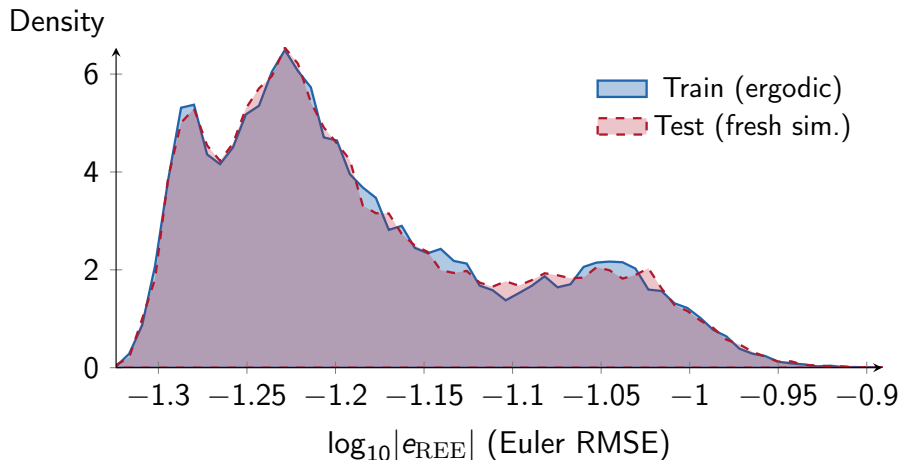
Backup: Slack Distribution by Size, 1993Q2–2019Q4

Distribution of Regulatory Slack by Bank Size



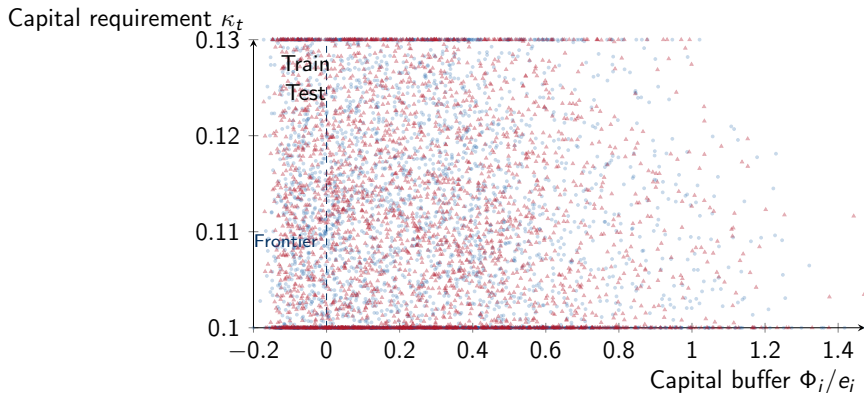
Shaded bands: P10–P90 (light) and P25–P75 (dark). Solid line: median.
Dashed vertical lines mark Riegle-Neal (1994Q4), GLBA (1999Q4),
Dodd-Frank (2010Q3, highlighted), and EGRRCPA (2018Q2).
Sample: quarterly bank panel, 1993Q2–2019Q4. Source: US Call Reports.

Backup: Euler Errors, Training vs. Test



Distribution of $\log_{10}|e_{\text{REE}}|$ on the training panel (seed 0) and an independent test panel (seed 42), 20,000 banks each after 200 periods of burn-in. Median residual -1.21 in *both* samples and no fattening of the test tail: the network generalises to unseen states rather than memorising the training trajectory.

Backup: Ergodic-Set Coverage



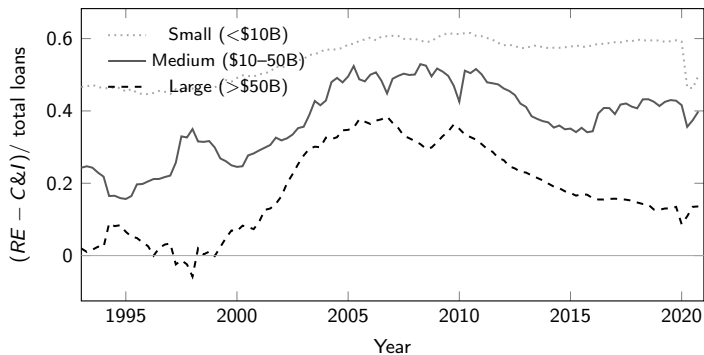
Joint distribution of the capital buffer Φ_i/e_i and the requirement κ_t , training against test simulation. The test cloud lies entirely inside the training cloud's convex hull, so the policy is evaluated only where it was solved — extrapolation beyond the training support is the most common failure mode of neural-network approximations.

Backup: Magnitudes in Context

Study	Design / setting	Effect of +1pp required capital
Aiyar, Calomiris, and Wieladek (2016)	UK bank-level requirement changes, 1998–2007	Lending growth 6.5–7.2pp lower
Mésonnier and Monk (2015)	EBA 2011–12 capital exercise, euro area	Annualised loan growth 1.2–1.6pp lower
De Jonghe, Dewachter, and Ongena (2020)	Pillar-2 decisions, Belgian credit register, 2013–15	Corporate credit growth ≈ 0.2 – 0.4 pp per quarter lower
Conti, Nobili, and Signoretti (2023)	Narrative BVAR, Italy (EBA/CA episodes)	Loan supply -1.5% after 2 years per +40bp Tier 1 ($\approx -3.8\%$ per pp, linearised)
This paper (model)	SMM-estimated policy, permanent +1pp κ	C&I -3.5% , RE -3.0% , total credit -3.1% (stationary comparison)
This paper (LPs)	Within-bank LPs scaled through the SMM mapping (one shock unit $\mapsto \hat{\sigma}_{\kappa} = 4.0$ bp of κ)	Trough responses $\approx -8.8\%$ (C&I, $h=16$) and $\approx -5.7\%$ (RE, $h=12$); out-of-sample linear extrapolation

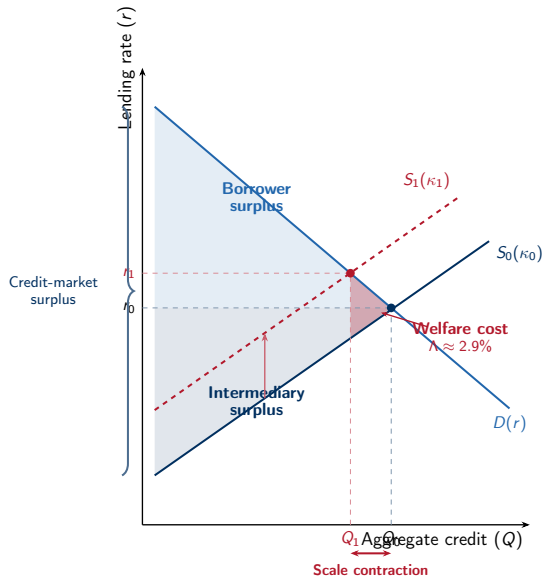
Model's -3.5% C&I / -3.0% RE / -3.1% total per +1pp sits squarely inside the quasi-experimental range. These magnitudes are conditional on the solved policy; the cross-sectional gradient is not.

Backup: Asset Gap



Difference between RE and C&I lending as a share of total loans, by size cohort, 1993–2020.

Welfare: Credit-Market Surplus



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